## B6 Stress, Strain and Young's Modulus

Preliminary question: A wire has a diameter of $1.6$ mm. What is its cross sectional area in m$^2$? Answer: $2.0 \times 10^{-6}$ m$^2$. If you can't see why this is true, ask your teacher before continuing with this section.

B6.1 Complete the questions in the table. The samples has a circular cross section.

| Diameter /mm | Cross sectional area /m$^2$ | Tension /N | Stress /MPa |
|---|---|---|---|
| $0.50$ | (a) | $4.9$ | (b) |

B6.2 A copper wire is put under tension. It was $1.80$ m long to begin with, and then stretches by $6.0$ mm. Calculate its strain.

B6.3 Copper has a Young's modulus of $120$ GPa. What is the largest diameter of copper wire we can use if a $1.80$ m length of wire is to stretch by at least $6.0$ mm under a tension of $20$ N?

## Answers

**B6.1** Area $= \pi \times \left(2.5 \times 10^{-4}\right)^2 = 1.9635 \times 10^{-7}$ m²

$$\text{Stress} = \frac{4.9}{1.9635 \times 10^{-7}} = 24.955 \text{ MPa}$$

**B6.2** $\dfrac{6 \text{ mm}}{1800 \text{ mm}} = 3.333 \times 10^{-3}$

**B6.3** Strain $\epsilon$ is answer to previous question

$$\text{Area} = \frac{F}{\sigma} = \frac{F}{E\epsilon} = \frac{20}{120 \times 10^{9}\epsilon} = 5.0 \times 10^{-8} \text{ m}^2$$

$$\text{Diameter} = 2 \times \sqrt{\frac{5.0 \times 10^{-8}}{\pi}} = 0.2523 \text{ mm}$$